Combustion and Flame 160 (2013) 1667–1676

Contents lists available at SciVerse ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

A PAH growth mechanism and synergistic effect on PAH formation in counterflow diffusion flames

Yu Wang [a], Abhijeet Raj [b], Suk Ho Chung [a,*]

[a] Clean Combustion Research Center, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia
[b] Department of Chemical Engineering, The Petroleum Institute, Abu Dhabi, UAE

ARTICLE INFO

Article history:
Received 8 December 2012
Received in revised form 10 March 2013
Accepted 11 March 2013
Available online 6 April 2013

Keywords:
PAH
Kinetic mechanism
Synergistic effect
Counterflow diffusion flame
Simulation

ABSTRACT

A reaction mechanism having molecular growth up to benzene for hydrocarbon fuels with up to four carbon-atoms was extended to include the formation and growth of polycyclic aromatic hydrocarbons (PAHs) up to coronene ($C_{24}H_{12}$). The new mechanism was tested for ethylene premixed flames at low (20 torr) and atmospheric pressures by comparing experimentally observed species concentrations with those of the computed ones for small chemical species and PAHs. As compared to several existing mechanisms in the literature, the newly developed mechanism showed an appreciable improvement in the predicted profiles of PAHs. The new mechanism was also used to simulate PAH formation in counterflow diffusion flames of ethylene to study the effects of mixing propane and benzene in the fuel stream. In the ethylene–propane flames, existing experimental results showed a synergistic effect in PAH concentrations, i.e. PAH concentrations first increased and then decreased with increasing propane mixing. This PAH behavior was successfully captured by the new mechanism. The synergistic effect was predicted to be more pronounced for larger PAH molecules as compared to the smaller ones, which is in agreement with experimental observations. In the experimental study in which the fuel stream of ethylene–propane flames was doped with benzene, a synergistic effect was mitigated for benzene, but was observed for large PAHs. This effect was also predicted in the computed PAH profiles for these flames. To explain these responses of PAHs in the flames of mixture fuels, a pathway analysis has been conducted, which show that several resonantly stabilized species as well as C_4H_4 and H atom contribute to the enhanced synergistic behaviors of larger PAHs as compared to the small ones in the flames of mixture fuels.

1. Introduction

Combustion generated polycyclic aromatic hydrocarbons (PAHs) and soot have been studied extensively because of their adverse health and environmental effects [1,2]. Especially, gaseous hydrocarbon fuels with up to four carbon-atoms (hereafter referred to as C_1–C_4 fuels) such as ethylene and propane have been studied in various flame configurations to understand the physical and chemical processes involved in PAHs and soot formation [3–7]. Detailed profiles for both major and minor species have been reported.

Appreciable effort has been devoted to developing and testing reaction mechanisms that can reliably predict the concentration of PAH molecules (which are used as soot precursors in soot models [1]). Frenklach and coworkers have proposed hydrogen-abstraction-acetylene-addition (HACA) mechanism for PAH growth [8–12]. Apart from acetylene, other species such as propargyl, allyl, cyclopentadienyl and indenyl have also been considered as contributors to PAH growth [13–23], mostly focusing on the role of resonantly stabilized radicals. The roles of phenyl addition/cyclization [24] and methyl addition/cyclization [25] have also been identified. The growth of large PAHs through coagulation has been modeled [26] where a detailed mechanism is used to describe the formation of aromatic species up to three rings [27] and coagulation takes place to form larger PAHs.

The PAH chemistry has also been developed and validated in [28–34]. In particular, PAH reactions channels present in the literature has been reviewed and further refined for C1 and C2 fuels for PAH growth up to five-ring aromatic species [32].

Such chemical mechanisms in predicting PAH formation has been tested for various flame configurations such as premixed flames [32,35–37], co-flow non-premixed [38–42] and counterflow non-premixed flames [33,43–47]. These mechanisms are typically limited to relatively small PAHs with up to four or five rings and were tested mainly for single component fuels.

Considering that larger PAHs are more likely to incept soot particles, a detailed description of their formation is important. Given that practical fuels typically include large number of hydrocarbons,

* Corresponding author.
E-mail address: sukho.chung@kaust.edu.sa (S.H. Chung).

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http://dx.doi.org/10.1016/j.combustflame.2013.03.013

it is important that a kinetic mechanism could be applicable to fuel mixtures. In this regard, a number of experiments have been conducted in counterflow diffusion flames (CDFs) of ethylene mixed with various hydrocarbons (binary component fuels), oxygen partial premixing, and oxygenated fuels [48–55] to study fuel structure effect on the formation and growth pathways of PAHs and soot.

Chemical cross-linking effects between binary component fuels observed in some of these flames have helped us achieve a clearer understanding about the underlying reaction channels. For instance, in the CDF with ethylene/propane mixtures, laser-induced fluorescence (LIF) and laser-induced incandescence (LII) techniques were used to measure relative PAHs and soot concentrations [48]. An enhancement in the productions of PAHs and soot as compared to the cases with individual fuel components was observed, revealing a synergistic effect. The synergistic effect was found to be more pronounced for larger PAHs as compared to small ones. This behavior has been explained based on the roles of: (a) C_2 species in PAH and soot growth through hydrogen-abstraction-C_2H_2-addition (HACA) mechanism, (b) methylene from methyl radicals in forming propargyl through reactions with C_2 species, and (c) propargyls producing incipient rings that can grow to form large PAHs.

Chemical reaction mechanisms developed for multi-component or surrogate fuels should be able to predict the above mentioned synergistic effects. However, computational simulations on such synergism were frequently limited to small aromatic species such as benzene because of the limitations in reliable reaction channels to large PAHs in many existing kinetic mechanisms. To the authors' knowledge, such synergistic effect, being more pronounced for larger PAHs as observed in [48], has never been tested through simulations.

A mechanism for gasoline surrogate fuels (referred to as KAUST PAH Mech 1 or KM1) [56] was developed from the base mechanism [57] to account for the growth of PAHs up to coronene ($C_{24}H_{12}$). KM1 has markedly improved the prediction of PAH concentrations as compared to the base mechanism. The base mechanism, however, was developed mainly for large hydrocarbon fuels. As such, sub-mechanisms for fuels with small hydrocarbons were simplified. This could lead to unsatisfactory predictions of the concentrations of small hydrocarbon species, some of which have been previously shown to be important for their synergistic effects [48,51]. Thus, KM1 may not be suitable in simulating flames of small hydrocarbon fuels to predict the behaviors of PAHs. It is therefore, useful to develop a kinetic mechanism for small hydrocarbon fuels that includes extended PAH chemistry containing PAHs with a wide mass range.

The objective of the present study was to develop a kinetic mechanism for C_1–C_4 fuels by including detailed reaction pathways for the formation and growth of PAH molecules up to coronene to enable the computation of the profiles of large PAHs. We validated this mechanism quantitatively with existing experimental data on premixed flames. And then, the mechanism has been applied to CDFs of ethylene/propane mixtures to test the prediction capability on PAH behaviors, such as the synergistic effect and the degree of synergism for different size PAHs together with the effect of benzene addition on PAH behaviors. Finally, the chemical cross-linking effect leading to the synergistic effects was analyzed.

2. Mechanism development

A recently updated and extensively validated mechanism for C_1–C_4 fuels [58] (hereafter called USC Mech) including molecular growth up to benzene was used as the base mechanism. This mechanism, containing 111 chemical species and 784 reactions, was extended by including reactions for PAH growth up to the formation of coronene based on KM1. The extended mechanism (called KM2) includes 202 species and 1351 reactions.

A brief description of notable features of KM2 is as follows. The role of cyclopentadienyl radicals (C_5H_5) in forming larger aromatics was described in [56,59–63]. Therefore, 17 reactions were included in USC Mech from [56] to improve the C_5H_5 sub-mechanism. These reactions mainly involve a pathway proposed in [59] for benzene formation from C_5H_5, formation of naphthalene through its recombination, and the abstraction of H atom from cyclopentadiene by H atom, methyl radical (CH_3) and propargyl radical (C_3H_3) along with the reverse reactions. H-abstraction reactions are particularly important because they activate chemical species for further reactions. In [61], the role of odd-carbon number species such as CH_3 and C_3H_3 in H-abstraction from PAHs and soot was discussed for diffusion flames. Thus, H-abstraction reactions for important species, such as benzene, PAHs and indene (C_9H_8) by H, CH_3 and C_3H_3 were also included. For all species smaller than C_5, the reactions and their rates were not altered from the base mechanism. In propane combustion, a noticeable amount of CH_3 is formed. Therefore, reactions involving the formation of benzene from C_5H_5 and CH_3, as proposed in [59], were included.

The reactions for the growth of PAHs larger than benzene are taken from [56]. The details are briefly reviewed here. Along with the HACA mechanism for PAH growth, several reactions involving species with odd-carbon number species such as indenyl (C_9H_7), benzyl ($C_6H_5CH_2$), C_5H_5 and C_3H_3 are included. The rate constant for the self-addition of C_5H_5 to form naphthalene were taken from an experimental study in [64]. For the prediction of the concentrations of cyclic C_5 species such as C_5H_5 and C_5H_6, reactions involving these species were added from [56] to the base mechanism. Some reactions involving the addition of C_4H_4 to large PAH radicals are also considered.

The rate constants for PAH reactions not present in the literature were determined through quantum calculations using the density functional theory (B3LYP hybrid functional and 6-311++G(d,p) basis set) along with the transition state theory. Note that none of the rate constants were arbitrarily changed or intuitively estimated to match the computed and the experimental results. Both forward as well as reverse rate constants were calculated for those reactions [56]. The thermodynamic data and the transport properties of all the chemical species present in KM2 were taken from [56,58], The rate constants for PAH reactions were obtained in the high pressure limit, as PAH molecules are large in size and their reactions do not exhibit substantial pressure dependence. The CHEMKIN format mechanism along with thermodynamic and transport data of species is included in the supplements. The chemical notation is the same as in [56] and aromatic species with 1–7 rings are named A1–A7, whose chemical structures are shown in Fig. 8 of this paper.

3. Results and discussion

3.1. Premixed flame simulations

Two premixed flames at different pressure conditions were simulated for the validation of KM2. First, the ethylene flame at 20 torr with $C_2H_4/O_2/Ar = 19.4/30.6/50.0$ was simulated (Flame 1). This flame was particularly chosen since experimental profiles for many intermediate C_1–C_6 hydrocarbons were available [65]. The simulation was conducted through the premixed flat flame model in Chemkin Pro software [66] using USC Mech [58], ABF Mech [11], KM1 [56] and KM2. Experimental temperature profiles were corrected for cooling effect of the probe (lower by 100 K and moved 0.5 mm away from burner surface as suggested in Ref. [65]) and then used as input for the simulation to avoid uncertainties in predicting flame heat transfer to the burner. Experimental concentrations were also shifted to the burner by 0.9 mm to account for probe effect [65] for the comparison with predictions. In the simulation, mixture-averaged transport properties were used with the correction velocity formalism to satisfy species mass conservation closure. Thermal diffusion effect (Soret effect) was considered in the species transport equations. Solution gradient and curvature were controlled to ensure smoothness of the calculated profiles and typically over 100 grid points were used.

The results on major species in Fig. 1 shows that all of the mechanisms tested are reasonably well predicted, although KM2 and USC Mech give better results for H_2 concentration. Regarding intermediate hydrocarbons, as presented in Fig. 2, KM2 shows improvements in the predictions for most species, especially those closely related to PAHs and soot formation such as CH_2 (triplet), C_2H_2 and benzene. The concentrations of CH_2 and C_3H_3 are over-predicted with KM2 and USC Mech. As will be shown later, a reaction pathway analysis revealed that C_3H_3 was mainly formed through the combination of CH_2 and C_2 species, and as such the over-prediction of CH_2 and C_3H_3 are related. However, given that these radical species are very reactive, it is reported that [67] there is an uncertainly (by a factor of 4) in their measurement such that over-prediction of their concentrations in the simulation may be acceptable. Note that the predictions of the peak positions of these radical species are improved by KM2. For C_3H_4, the computed concentrations of propadiene and propyne were summed together to compare with the experimental data as the experimental technique used in [65] was not able to distinguish between these isomers. C_4H_4 in the mechanism represents vinylacetylene.

The improvement in the prediction of benzene concentration with KM2 compared to KM1 and ABF Mech is seen to be appreciable. This is promising considering the vital role of benzene in forming PAHs. Note that for most species, KM2 and USC Mech give nearly identical results. This is as expected since the sub-mechanism of KM2 for molecular growth from fuel to benzene was taken from USC Mech with only slight modifications, as mentioned in the previous section. This implies that the improvement in the predictions with KM2 as compared to KM1 regarding C_1–C_6 species is largely due to the incorporation of USC Mech. The well prediction of benzene is one of the motivations for us to choose USC Mech as the base mechanism.

The prediction of large PAHs is a focus of this paper such that an additional ethylene premixed flame with the equivalence ratios of 3.06 at the atmospheric pressure (Flame 2) was simulated, where experimentally measured PAH concentrations are available [36]. Effect of gas exit temperature on exit velocity was considered to ensure that the reactant mass fluxes (cold gas velocity) equal the experimental values [36]. Figures. 3 and 4 show the comparison of experimental and computed concentrations of major products, benzene and PAH molecules. The simulation results from USC Mech, ABF Mech and KM1 were included for comparison when applicable. Note that Flame 2 is a sooting flame such that soot formation influences PAH profiles, while the present simulations did not consider a soot model. However, the PAHs shown in Fig. 4 are relatively small ones and have boiling temperatures lower than 500 °C, which are less likely to stick together at flame temperatures to contribute significantly to the nucleation of soot particles [2].

Most of major gaseous products were well predicted, except that C_2H_2 concentration was somewhat underpredicted by the tested mechanisms. Considering the performance of KM2 and USC Mech in C_2H_2 predictions in Flame 1 (Fig. 2) and the fluctuation in the experimental values, measurement uncertainties might also contribute to this discrepancy. In fact, the under-prediction of C_2H_2 is also seen by simulation with other mechanisms [14,33,36]. When comparing KM2 and USC Mech, the predicted C_2H_2 profile with KM2 is lower than that with USC Mech, as compared with the case shown in Fig. 1 where both mechanisms give nearly iden-

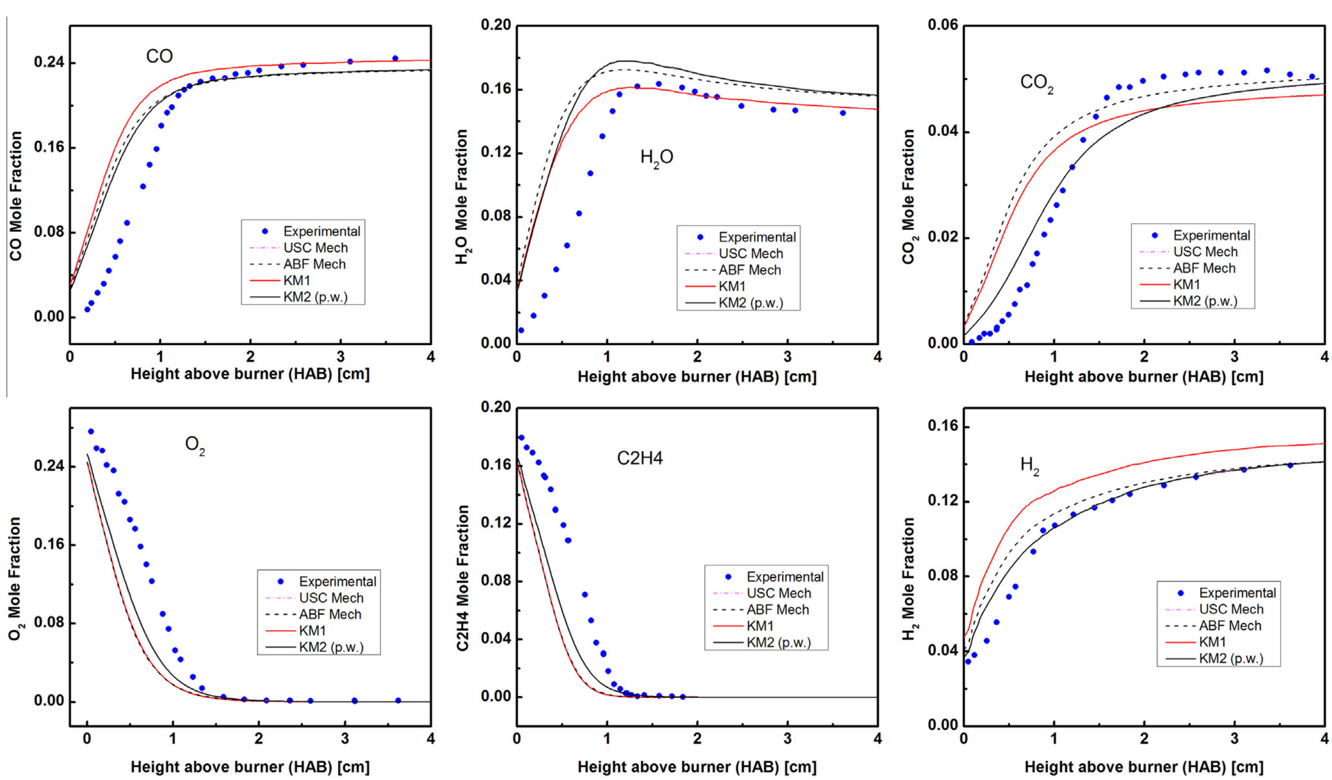

Fig. 1. Comparison between experimental and calculated mole fraction profiles of major species in C_2H_4 premixed flame [65] (20 torr, $C_2H_4/O_2/Ar$ = 19.4/30.6/50.0). The results from KM2 and USC Mech overlap for most of the species.

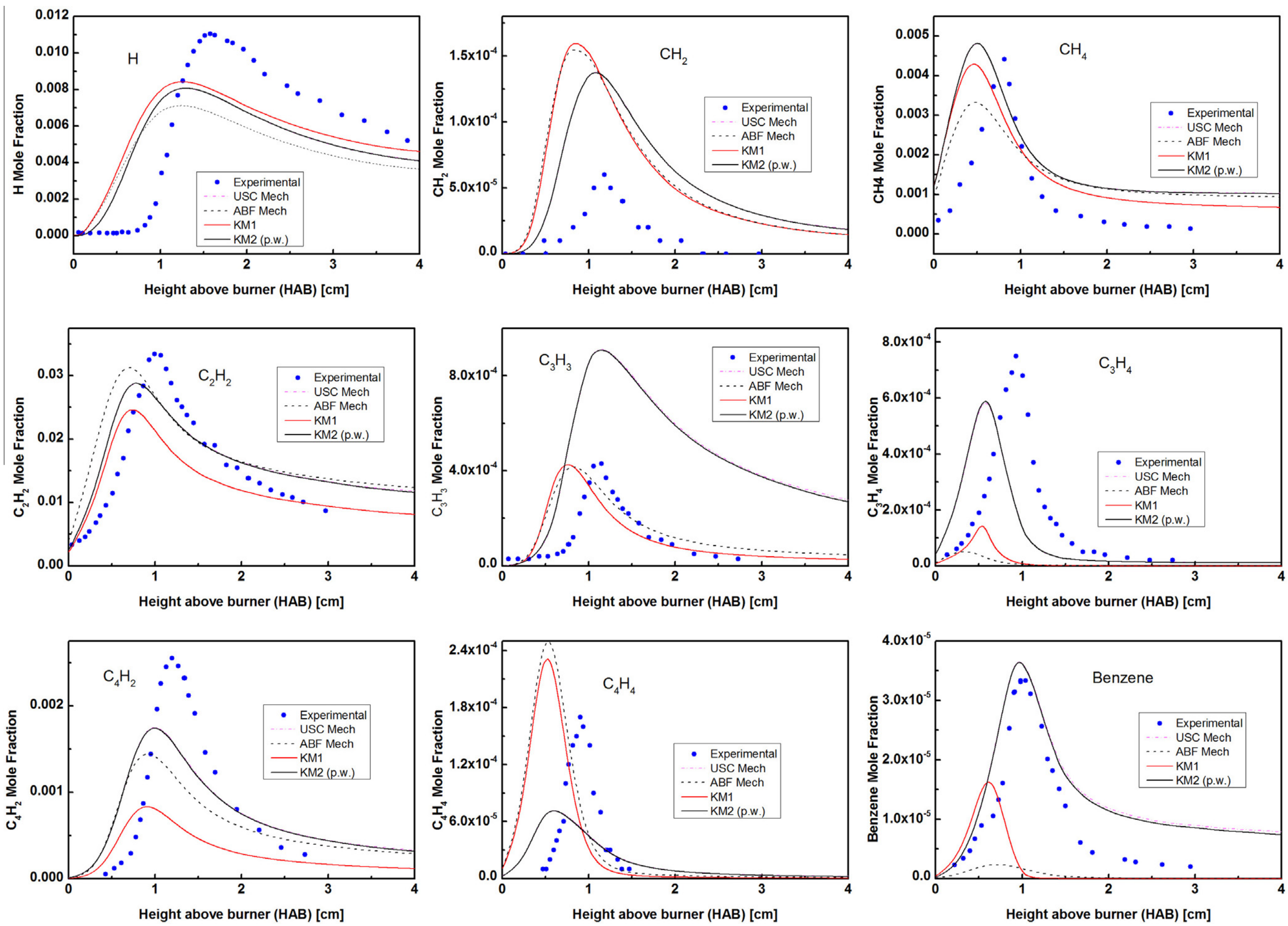

Fig. 2. Comparison between experimental and calculated mole fraction profiles of minor species in C_2H_4 premixed flame [65] (20 torr, $C_2H_4/O_2/Ar$ = 19.4/30.6/50.0). The results from KM2 and USC Mech overlap for most of the species.

tical results. This can be attributed to the consumption of C_2H_2 to form PAHs in Flame 2, which has higher PAHs and soot loading as compared to Flame 1 due to higher equivalence ratio and pressure. Note that USC Mech does not account for PAH formation. The prediction of benzene and most other PAH concentrations in Flame 2 was improved with KM2 as compare to KM1, as shown in Fig. 4, except A4.

Significant improvements can be seen in Fig. 4 in the predicted concentrations of large PAH molecules by KM2 and KM1 as compared to the results from ABF Mech, which is mainly based on the HACA mechanism. For instance, ABF mechanism under-predicts the A3 concentration by a factor of more than 10 and A4 concentration by a factor of more than 100 (a linear scale version of Fig. 4 is also provided in supplementary material). The concentration of A4R5 from ABF Mech is not available because it was not included in the simulation. These results emphasize the importance of the chemical pathways other than the HACA mechanism, involving odd carbon number species for PAH formation and growth even in the flames of small hydrocarbons such as C_2H_4. These pathways are included in KM2 as described previously. The use of KM2 to study the behaviors of PAHs in counterflow flames is thereby warranted.

3.2. Counter-flow diffusion flame simulations

In Ref. [48], experiments were conducted on the counterflow diffusion flames of ethylene–propane mixtures. Relative PAH concentrations from PAH LIF measurements were provided. The oxidizer stream of these flames had 76% N_2 and 24% O_2 and the fuel stream was composed of a mixture of C_2H_4 and C_3H_8 diluted with 27% N_2. The nozzle exit velocities U_0 were 20 cm/s with the separation distance between the nozzles being 1.42 cm. The propane ratio (R_P) was defined as the ratio of the volumetric flow rate of propane to that of the total fuel.

$$R_P = \frac{Q_{C_3H_8}}{Q_{C_3H_8} + Q_{C_2H_4}}$$

where Q represents the volumetric flow rate. The effect of change in R_P on relative PAH behaviors was examined. PAH LIF signals were measured at different detection wavelengths. These are soot formation (SF) flames [68], which are located on the oxidizer side of the stagnation plane. The PAH and soot formation zones are on the fuel side of the flame and the convective field tends to transport soot particles generated from large PAHs away from the flame zone without intensive soot oxidation. Thus, these SF flames are well suited for our study of PAH growth. Note, however, there are also similar experimental studies on soot formation and oxidation (SFO) flames [69] where PAH and soot oxidation is of importance.

Synergistic effects were observed in these signals as R_P was varied from 0 to 1. The degree of synergism tended to be higher for signals at higher detection wavelengths. Since LIF signals at higher detection wavelengths are believed to represent larger PAH molecules [70,71], the experimental observations show that larger PAHs have higher degrees of synergistic effects. We used KM2 to simu-

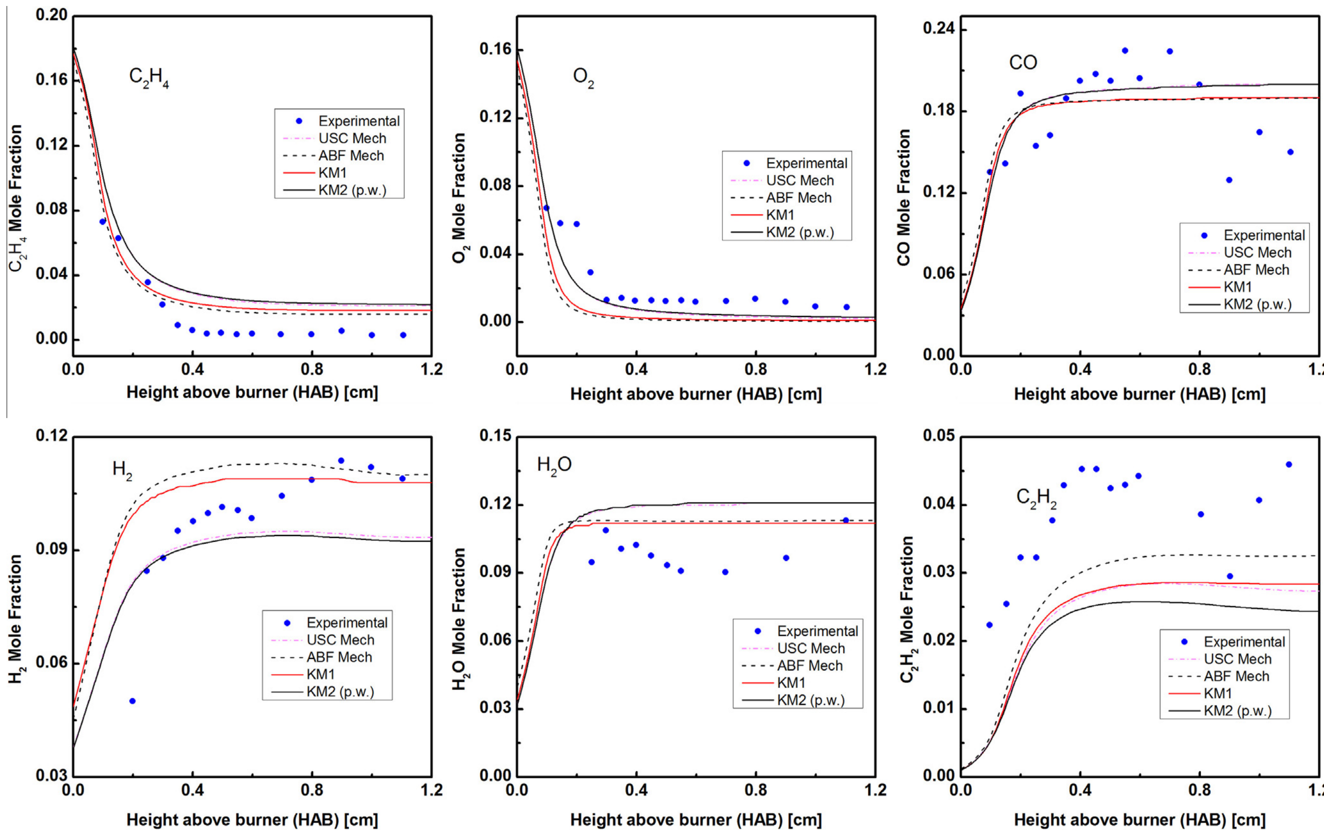

Fig. 3. Comparison between experimental and calculated mole fraction of major gaseous products in C_2H_4 premixed flame [36](1 atm, $C_2H_4/O_2/Ar$ = 21.3/20.9/57.8).

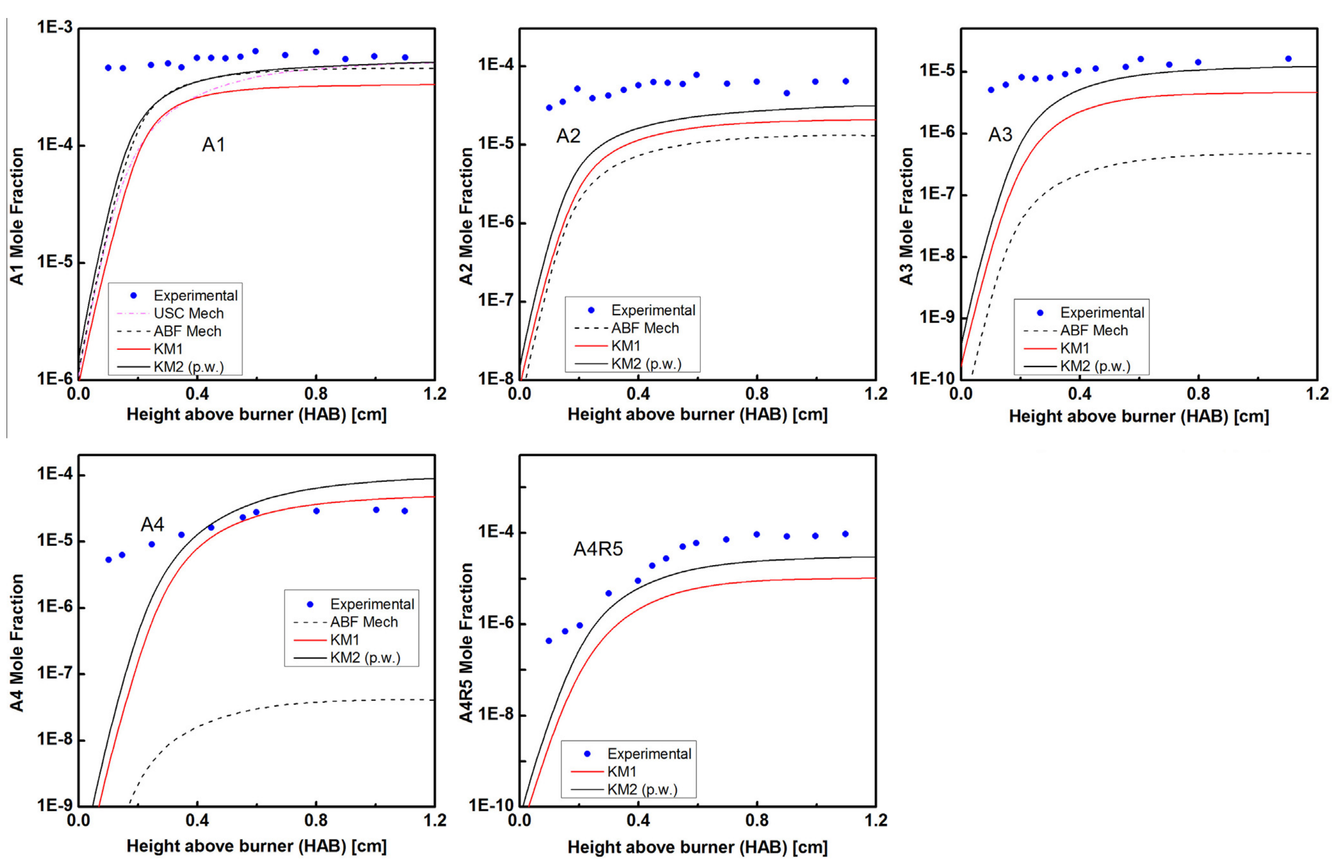

Fig. 4. Comparison between experimental and calculated mole fraction of benzene and various PAHs in C_2H_4 premixed flame [36] (1 atm, $C_2H_4/O_2/Ar$ = 21.3/20.9/57.8).

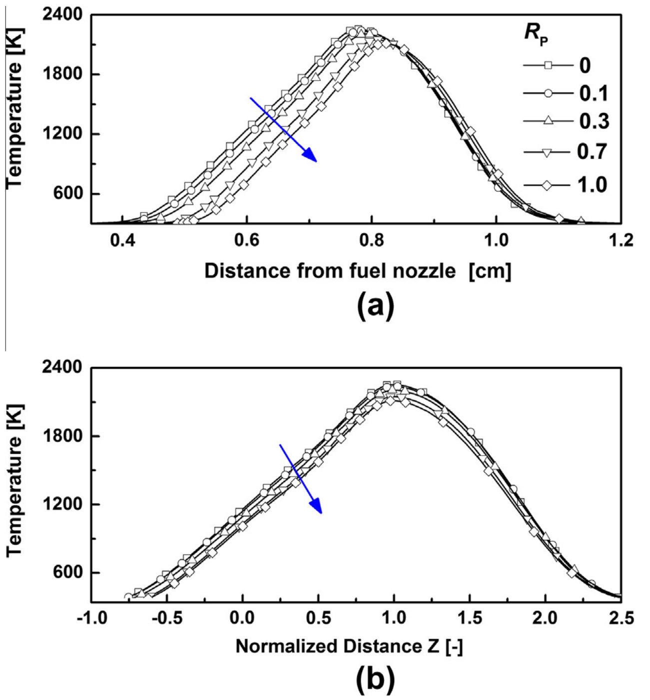

Fig. 5. Calculated temperature profiles of the counterflow flame at several R_P (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s) (a) as a function of physical distance (b) as a function of normalized distance.

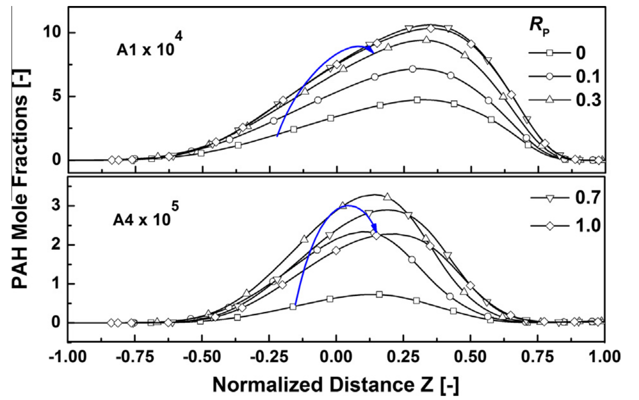

Fig. 6. Calculated PAH profiles of the counterflow flame at several R_P (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s)

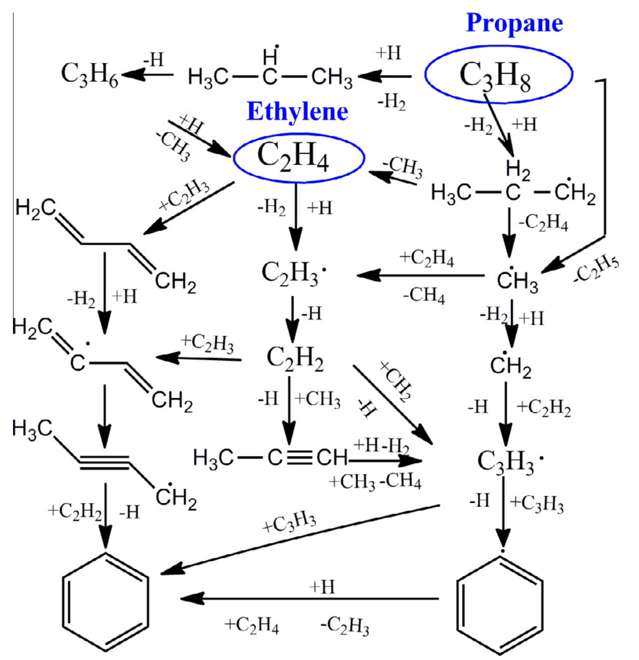

Fig. 7. Kinetic pathways from fuel mixtures (R_P = 0.1) to benzene (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s, for region $-0.5 < Z < 0.75$).

late these experiments. It will be shown later that our simulation results agreed well with the experimental trends regarding synergistic effects on PAH formation. Before that, the flame structures of some of the simulated flames and the pathways leading to the formation of benzene and large PAHs in the flames of fuel mixtures are presented.

Figures 5 and 6 provide the calculated temperature profiles and A1 and A4 concentration profiles with varying R_P, respectively. The temperature and aromatic species profiles shift towards the oxidizer side as R_P increased, which is due to the shift of the flame position caused by the varying stoichiometry as R_P changed. Since the locations for the formation of aromatic species are dependent on the flame position, the profiles were also plotted against normalized distance, Z, defined as $Z = (X - X_{st})/(X_{T,max} - X_{st})$, to highlight the chemical effects on PAH profiles [50], where X and T are the distance from the fuel nozzle and temperature, respectively, and the subscripts 'st' and 'max' indicate the stagnation plane and maximum, respectively. Note that the blue arrows indicate the increasing trend of R_P.

The result shows that the maximum temperature decreases monotonically, while the peak A1 and A4 mol fractions change non-monotonically as R_P is increased. The decreasing temperature can be attributed to the variation in stoichiometry through nitrogen dilution with R_P. The non-monotonic variation in the concentrations of aromatic species (synergistic effect) is not solely due to thermal effects since the flame temperature changes monotonically. Additional calculations (although not shown here) show that the increase in the nozzle exit temperature at fixed R_P resulted in monotonic increase in PAH concentrations. The chemical 'cross-linking' between the mixed fuels is thus believed to be responsible for the observed PAH synergistic effect, which is elaborated below.

Figure 7 shows the important reaction paths from the two fuels of ethylene and propane leading to A1 (benzene) obtained from rate of production analysis (ROP) in Chemkin Pro [66], which identifies the pathways of the formation of aromatic species. ROP was conducted for R_P = 0.1 and evaluated at various flame positions ($-0.5 < Z < 0.75$) and those reactions contributing most were selected to be shown. The self-addition of propargyl (C_3H_3) and the addition of acetylene to C_4 species are found to be the important reactions for benzene formation. The main pathway to C_3H_3 formation is the reaction between C_1 and C_2 species. Note that when a small amount of propane is present, most of the C_1 radicals are formed through propane decomposition. C_2 species mainly come from the dehydrogenation of ethylene. As R_P increases, C_2H_2 concentration decreases. This opposing trend of C_1 and C_2 concentrations contributes to the non-monotonic variation of C_3H_3 and benzene concentrations. Furthermore, since benzene is one of the building block for large PAH molecules and odd-carbon number species reactions are also necessary for PAH growth, it is reasonable to expect their synergistic effects to propagate to large PAH molecules.

Figure 8 provides the reactions for the formation of PAHs ($-0.5 < Z < 0.75$). The reactions involving PAH formations from their radicals through H addition were omitted to simplify the diagram. Clearly, apart from the HACA mechanism, several reactions

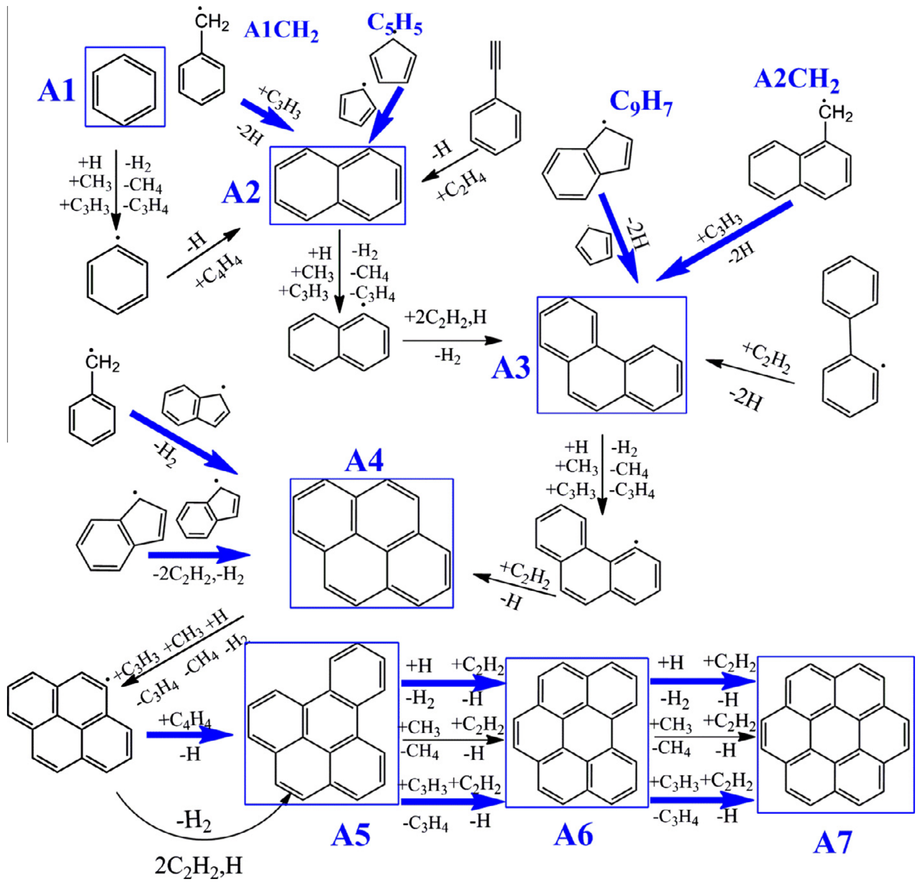

Fig. 8. Kinetic pathways from benzene to coronene (A7) for fuel mixtures ($R_P = 0.1$) (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s, for region $-0.5 < Z < 0.75$). A1: benzene, A2: naphthalene, A3: phenanthrene, A4: pyrene, A5: benzo[e]pyrene, A6: benzo[ghi]pyrene, A7: coronene.

involving odd-carbon number species such as C_9H_7 and C_5H_5 contribute to the PAH growth in the flames of ethylene–propane mixtures. In addition, hydrogen abstraction through C_3H_3 is included to form active sites on PAH molecules for further reactions. Note that blue arrows are used to highlight those reactions contributing to the synergistic effects on PAHs. The roles of the reactions involving odd-carbon number species in the synergistic effect will be discussed later.

To illustrate the synergistic effects in PAHs, Fig. 9 compares the experimental peak PAH LIF intensity at four different detection wavelengths ranging from 330 to 500 nm [48] with the computed

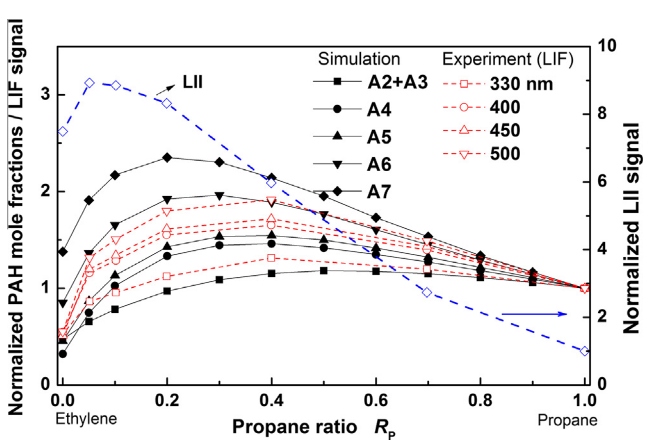

Fig. 9. Comparison of normalized PAH mole fractions and LIF signals (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s, experimental data from [48]). Absolute values of mole fraction for A2–A7 at $R_P = 1$ are A2: 5.66E−5, A3: 2.93E−5, A4: 2.28E−5, A5: 1.39E−6, A6: 1.79E−6, A7: 1.19E−5.

peak mole fractions of PAHs of different sizes. All the data are presented in terms of R_P and normalized by the propane case ($R_P = 1$). Since PAH LIF signals detected at 330 nm when excited at 266 nm were reported to represent two- or three-ring aromatic species [70,71], the peak mole fractions for A2 and A3 are therefore combined for comparison. As the detection wavelength increases, the LIF signals originate from larger PAH molecules [48]. The simulated peak profiles for large PAHs (A4–A7) are also presented.

As can be seen in the simulation results, the degree of synergism becomes more pronounced for larger PAH molecules, which agrees qualitatively with the experiments. Note that the simulated result for A7 is higher for $R_P = 0$ than that for $R_P = 1$, while the experimental LIF signal always shows a higher intensity for $R_P = 1$. This discrepancy can be understood by the difference in soot loading at various R_P, as confirmed by the LII signals included in Fig. 9, which is indicative of soot volume fraction. As mentioned previously, the simulation results presented here do not include any soot model. Experimental LII signals showed that the soot loading for pure ethylene is much higher than for propane. Since PAHs are precursors for soot and they are consumed by nucleation and condensation in soot formation process, it is reasonable to expect the calculated concentration of large PAHs to decrease more for pure ethylene than for pure propane. This implies that the agreements between the simulation and the experimental results for large PAHs will become quantitatively better if soot model is included in the calculation, which will be our future study.

The more pronounced synergistic effect for larger PAHs implies that PAH growth rates are affected by other synergistic factors besides the formation of incipient rings. This was further validated by examining the variation in PAH concentrations with R_P in flames of ethylene–propane mixture, where the fuel stream was doped with

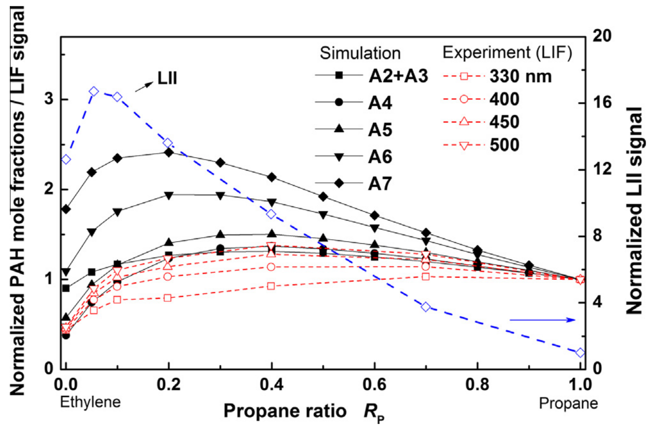

Fig. 10. Comparison of normalized PAH mole fractions and LIF signals for benzene doping cases (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, 1.83% benzene doping, U_0 = 20 cm/s, experimental data from [48]). Absolute values of mole fraction for A2–A7 at R_P = 1 are A2: 7.9E−5, A3: 3.86E−5, A4: 3.1E−5, A5: 2.04E−6, A6: 2.64E−6, A7: 1.76E−5.

benzene [48]. Figure 10 compares the simulated and experimental results for the case when 1.84% benzene was added to the $C_2H_4/C_3H_8/N_2$ fuel stream. As can be seen, the experimental trend for PAH was captured by the simulation. Since benzene is added in the fuel, its concentration in the flame is much higher than in the non-doping case and as such its formation from small hydrocarbons is expected to be less important. This could suppress the synergistic effects on benzene formation (as most benzene is now directly from the fuel). This can be seen in the predicted benzene profile as shown in Fig. S2 in supplementary material. Experimentally, synergistic effects in LIF signals at 330 nm are much weaker compared to those for longer detection wavelengths. However, the synergistic effect is still significant for larger PAHs, as can be seen in both computed as well as experimental results. This is not likely due to the synergistic behavior in the formation of incipient rings.

From another point of view, the successful prediction of synergistic effects even with abundant amounts of benzene addition qualitatively validates the chemical pathways in KM2 involving odd-carbon number species for the growth of large PAHs. In the simulation results the synergistic effects for smaller PAH are mitigated compared to the case without benzene doping, as expected. The agreement between the simulation and experimental results is qualitatively satisfactory, but the degree of synergism differs quantitatively. This again could be partly attributed to the enhanced soot formation caused by benzene doping and the exclusion of soot model in the current simulation. Experimental results show that the ratio of soot loading for ethylene to that for propane is even higher in the case with benzene addition compared to the case without benzene addition (as indicated by the LII signals). Higher soot loading leads to greater discrepancies between the simulation and experimental results as previously mentioned.

The degree of synergism was observed to be higher for larger PAHs as compared to the small ones. For better understanding of this enhanced synergistic effects for large PAHs, the variations in the concentrations of species that are involved in PAH growth (shown in Fig. 8) with R_P were studied. The concentration of C_2H_2, which is involved in PAH growth through the HACA mechanism, was found to decrease monotonically as R_P was increased (not shown here). Thus, it is not likely to contribute to PAH synergistic effects. In Fig. 11, the mole fraction profiles of several odd-carbon number species are plotted against the normalized distance Z to avoid complications from flame position variations at different R_P. The profiles are shown for the regions relevant to PAH formation ($0.1 < Z < 0.3$). At a specified Z, synergistic effects in the concentrations of these species are observed (indicated by the blue arrows and shown in the subfigure at the right). For example, the mole fraction of $C_6H_5CH_2$ and C_9H_7 at R_P = 0.3 are higher than those for pure fuels (R_P = 0 and 1). The synergistic effects on larger PAHs are thus expected considering the involvement of these species in

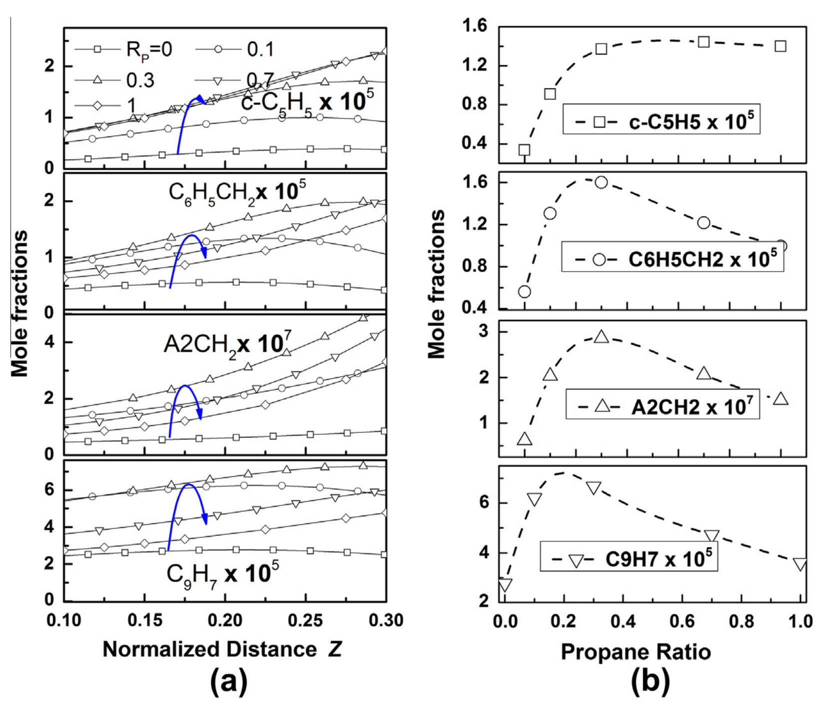

Fig. 11. (a) Mole fraction profiles for odd-carbon number species in PAH formation region (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s), (b) Mole fractions of odd-carbon number species for various propane ratios at Z = 0.2

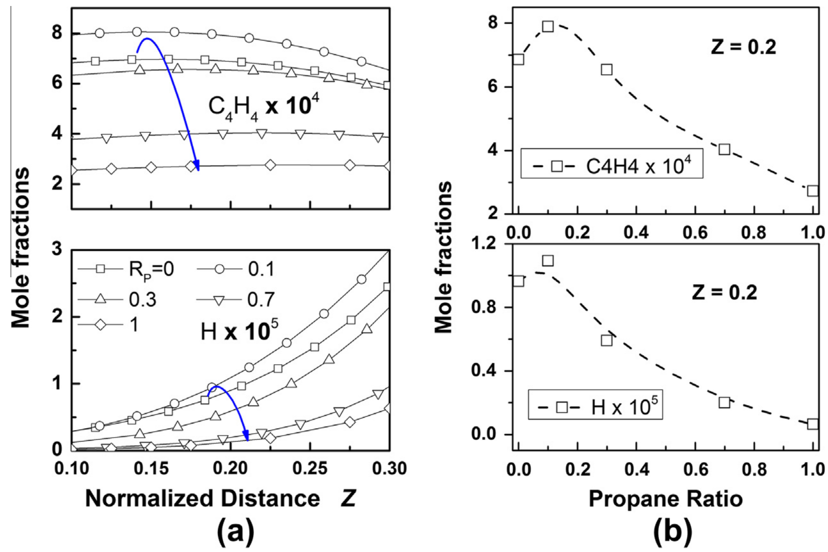

Fig. 12. (a) Mole fraction profiles for H and C_4H_4 species in PAH formation region (1 atm, Fuel/N_2 = 73/27, O_2/N_2 = 24/76, U_0 = 20 cm/s). (b) Mole fractions of C_4H_4 and H for various propane ratios at a Z = 0.2

forming large PAHs, as shown in Fig. 8, where the pathways that can lead to synergistic effects were marked with blue arrows.

Furthermore, the non-monotonic behaviors of these odd-carbon number species also help to explain the more pronounced synergistic effects for larger PAHs. The synergistic effect of benzene is seen to be mainly due to C_3H_3 which also behaves synergistically in ethylene/propane flames [51]. However, for large PAHs, multiple reaction channels contribute to their synergistic behaviors. For example, A2 can be formed through C_3H_3 reacting with $C_6H_5CH_2$ and through C_5H_5 self-addition. Chemical pathways leading to the formation of A3 include C_9H_7 reacting with C_5H_5 and $A2CH_2$ with C_3H_3. A4 can be formed through self-addition of C_9H_7 and its reaction with $C_6H_5CH_2$. Figure 11 shows that the reactants for these reactions, C_3H_3, C_5H_5, $C_6H_5CH_2$, $A2CH_2$ and C_9H_7, exhibit the synergistic effect. It is therefore reasonable to expect this effect to propagate and be enhanced for larger PAHs formed through multiple reaction channels involving above-specified species.

Note that the species shown in Fig. 11 are resonantly stabilized radicals (RSRs), which are relatively stable and can be present in the high temperature region for PAH growth. Reactions for the formation of PAHs beyond A4 from odd-carbon number species are not present in the mechanism, and they mainly grow through the addition of C_2H_2 and C_4H_4 to their radicals. In the formation of their radicals, H abstraction reactions by CH_3, C_3H_3 and H species are important. In addition to C_3H_3, which was discussed above, synergistic behaviors were also found for H and C_4H_4, as shown in Fig. 12. For these two species, the peak concentrations appear at a lower R_P (0.3) compared with the RSRs shown in Fig. 11. This qualitatively explains the shift in the peak position to lower R_P as the PAH size increases, as shown in Figs. 9 and 10.

In Figs. 9 and 10, the PAH profiles for large PAH are generally higher than the small PAHS, which indicates that the synergistic effect is more pronounced in large PAHs. In addition, the propane ratio at which the PAH concentration peaks also shifts to smaller value as the PAH size increases. Interestingly, the profiles of PAHs (against R_P) approach the LII profiles as their size is increased on these diagrams. Since LII signals represent the concentration of soot particles in flames, this may imply that larger PAHs could be involved in soot nucleation, and they should be used in soot inception models. However, all the existing models frequently consider soot inception from PAHs with up to 4–5 rings only, and thus require modifications.

4. Concluding remarks

A reaction mechanism for C_1–C_4 fuels is extended by including reactions for the formation and growth of PAHs up to coronene. The new mechanism is validated in several premixed flames, where a reasonable agreement between the observed and simulated PAH concentrations is obtained.

The synergistic effect is an important phenomenon that has been observed experimentally in several counterflow flames of fuel mixtures. The new mechanism was used to predict this effect by simulating the counterflow flames of ethylene–propane mixtures, where the PAHs with one to seven aromatic rings are shown to exhibit synergistic behaviors. In experiments, as the detection wavelength of LIF signals increased (i.e., increasing PAH mass), the degree of synergism was found to increase. This trend was captured in the simulated PAH profiles. It has also been shown through the simulation of the flames of ethylene–propane mixtures doped with benzene that the synergistic effect for larger PAHs prevails even with the addition of benzene, as observed experimentally. The role of species with odd-carbon number species such as C_1 and C_3 has been shown to be important in leading to the synergistic effects on PAHs through the rate of production analysis.

Several highly stable RSRs such C_5H_5, $C_6H_5CH_2$, $A2CH_2$ and C_9H_7 with odd-carbon number species as well as C_4H_4 and H atoms have been identified to be responsible for the enhanced synergistic behaviors of larger PAHs as compared to the small ones in the flames of fuel mixtures.

Acknowledgment

This work is supported by Saudi Aramco through KAUST CCRC.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2013.03.013.

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